Perspectiva

COM-480: Data Visualization
Milestone 2

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Sketches of visualization & core idea

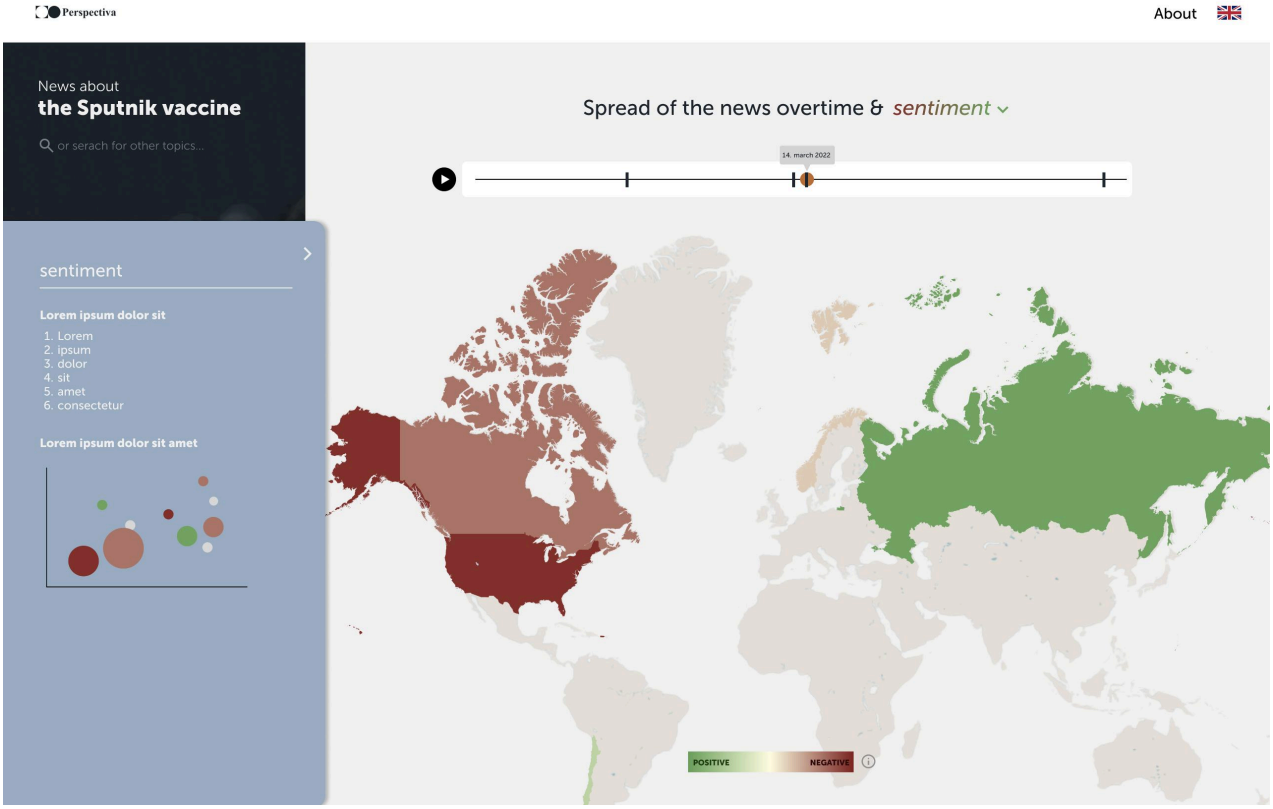

Figure 1: landing page prototype sketch made in figma.

Figure 1: The core visualisation is a 2D world map where countries are coloured by sentiment and news volume as the timeline progresses. Users will land directly into the main visualisation. After selecting a news topic (for this milestone: "the Sputnik vaccine"), the map becomes the central focus for the user. The user can choose to play and pause the timeline, and watch how the map evolves over time. The timeline is bounded by the natural news cycle of each topic. Rather than using fixed dates, the start and end points are determined dynamically, and the timeline begins when coverage picks up and ends when it drops below a set threshold, for example fewer than 10 articles published within a given period.

The vertical lines on the timeline represents tipping points/important events, that gives the user context to major changes. This will show if the user hovers over these marks, as well as showing up as a pop up window on the right side of the webpage with a topic and a small descriptive text.

The primary visualisation layer will be sentiment over time. This is the first thing users see and the “default” of the webpage. They can switch to other layers, such as news volume (colour intensity reflecting article count, as shown in Figure 2).

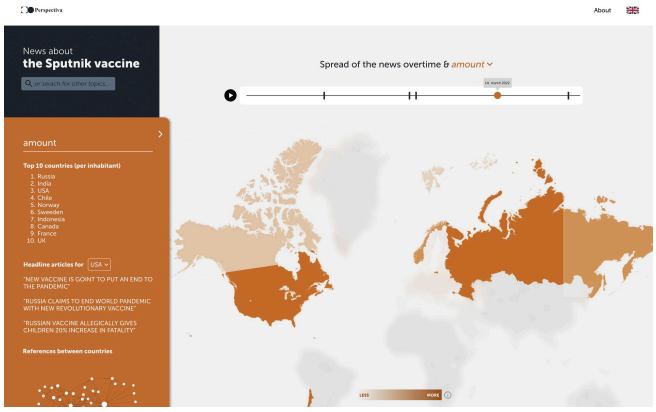

Figure 2: prototype sketch made in figma.

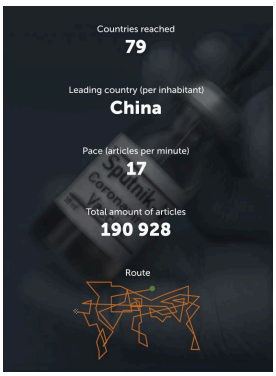

Figure 3: additional visualizations

On the left, a side panel will provide qualitative context and additional visualisation such as for example: the top 10 countries by article count, key headlines for a chosen country, and additional visualisations outlined in the "list of extra ideas" section.

The site will follow a martini glass structure: the map serves as the broad entry point, narrowing into a scrollable storyline with curated insights, shown on the left side panel. The idea here is to show meaningful patterns that can be translated across different news topics.

List of extra ideas

As described, the core idea is to show how news spread in the world over time. If time permits, additional topics will be added, such as “the Ukranian war”, “the 2008 financial crisis” etc. It is important that these other topics fit well into the original format, so that the users can analyse the same things across the different topics.

Beyond expanding the topic selection, we also want to implement an interactive scrollable storyline that provides deeper analysis. Ideally, the structure of this storyline will be meaningfull regardless of the topic. It has to be a narrative format that translates well across different topics.

In addtion to this, we would like to explore adding other layers to the map beyond sentiment and volume. One idea for this is connections between articles, so that users can se the “DNA” or family tree of events and topics. This would also show which countries that typically funcitons as “superspreaders” of news. Other layers that can be added remains to be determined.

Tools

- Next.js (React) & Tailwind CSS
- MapLibre GL & react-map-gl
- Google BigQuery
- Python (Pandas & PyArrow)
- GitHub Pages & Actions
- Figma

Lectures

Lectures Course material directly informed our prototype and will guide our final build:

Past lectures applied to our prototype:

- Interactions (Mar 16–22): Built the timeline playback, sliding detail panel, and dynamic tooltips.
- Perception of colors & Marks/Channels (Mar 23–29): Mapped sentiment to color gradients and accurately scaled our bubble charts based on article volume.
- Designing viz / Do & Don't (Apr 13–19): Structured our UI layout, visual hierarchy etc.

Future lectures:

- Maps & Practical maps (Apr 20–26): Will help optimize our MapLibre implementation and geospatial rendering.
- Storytelling (May 18–24): Will frame the final user experience